

PROPOSED SOLUTION: PETDOC – AN INTEGRATED AI-DRIVEN VETERINARY ECOSYSTEM

Course: CSE299
Section: 04
Group: 05

Group Members:
2312559042 – Rishat Bin Sultan
2232239642 – MD Tamim Hasan
2311101642 – Fariha Islam
2311677042 – Nafia Haque Muskan

Abstract— The PetDoc system provides a comprehensive technical solution to the inefficiencies of current veterinary healthcare by integrating advanced Machine Learning with a functional e-commerce framework. To address geographical and time constraints, the platform utilizes a Convolutional Neural Network (CNN) architecture for instantaneous, image-based diagnostics, offering pet owners immediate health guidance without the need for physical travel. Unlike previous research that we have seen in Literature Review that remained theoretical or focused on administrative tasks, PetDoc deploys a real-world, Django-based platform that unifies symptom analysis, medical procurement, and 24/7 expert consultation into a single interface. By bridging the "Research-to-Application" gap, the system creates a continuous healthcare pipeline that ensures early disease awareness, streamlined access to medicines, and constant professional oversight in one cohesive ecosystem.

I. Problem Definition and Proposed Strategy

The veterinary sector faces systemic barriers that lead to delayed care and high animal mortality. Pet owners are often restricted by geographical distance to clinics and the prohibitive costs of emergency visits, causing them to delay medical intervention until a condition becomes critical.

However, the core issue is the absence of accessible preliminary diagnostic tools. Without a reliable way to assess health at home, owners rely on generalized internet searches that often provide misinformation. PetDoc solves this by providing an immediate, AI-backed first line of defense. By utilizing Machine Learning, the platform provides instant, data-driven health guidance, empowering owners to identify urgent risks and seek treatment immediately, effectively bridging the gap between the first sign of illness and clinical care. Ultimately, the solution fosters a proactive healthcare environment where data-driven insights and essential products are available 24/7, significantly reducing the risks associated with untreated or late-diagnosed ailments.

II. ADDRESSING RESEARCH GAPS AND SOLUTION

Based on the analysis of existing literature (Hossain et al., 2020; Kumar & Singh, 2021), several "lacking" areas were identified. PetDoc is engineered to fulfill these specific deficiencies:

1. **From Theoretical Models to Functional Deployment:** Most current research focuses solely on the accuracy of ML models in a lab setting. PetDoc fulfills this gap by deploying these models into a real-world web application (Django) where pet owners can actually use them.
2. **Fragmented vs. Unified Systems:** Existing solutions are often "Fragmented"—platforms either offer e-commerce or simple record management. PetDoc fulfills the "Missing Piece" by combining Diagnosis + Treatment (E-commerce) + Professional Advice in one unified interface.
3. **Human-Centric to User-Centric Design:** While many systems are designed for clinic administrators, PetDoc is designed for the pet owner, ensuring that the interface is accessible to non-professionals as well.

III. TECHNICAL ARCHITECTURE OF THE PROPOSED SOLUTION

The solution is divided into three interconnected modules:

A. AI-Based Disease Detection Module

To solve the lack of automated preliminary guidance, PetDoc implements a dual-input diagnostic engine:

- **Image-Based Detection:** Utilizing Convolutional Neural Networks (CNN), the system processes user-uploaded images of skin irritations, eye infections, or

visible injuries. The model extracts structural features to categorize the severity and type of the ailment.

- **Symptom-Based Classifier:** For internal or behavioral issues, a Random Forest or Support Vector Machine (SVM) classifier processes text-based inputs (e.g., vomiting, lethargy) to provide a list of potential conditions.

B. Integrated Veterinary E-Commerce

Addressing the gap in medical supply chains, the platform includes a full-stack e-commerce engine:

- **Contextual Recommendations:** Unlike generic pet stores, PetDoc suggests products based on the AI's diagnostic output. If a skin condition is detected, the system highlights medicated shampoos or ointments that is generated through a link that send the pet owner to the e-commerce platform.
- **Secure Transactions:** Built on the Django framework, it ensures secure payment processing and order tracking for medicines and nutritional essentials.

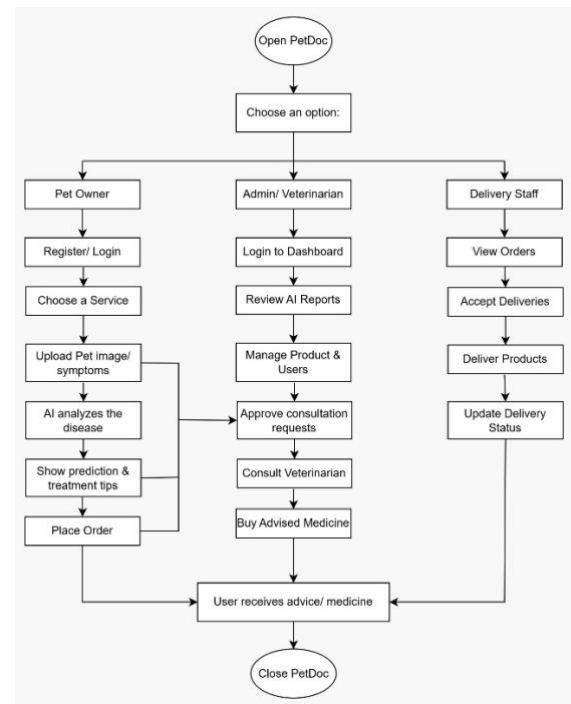
C. 24/7 Virtual Support Layer

To ensure that the platform is both safe and reliable, PetDoc includes a 24/7 consultation service. This feature allows users to escalate their concerns to a certified veterinarian via a live dashboard, bridging the gap between digital "self-help" and professional medical intervention.

IV. SYSTEM FEASIBILITY AND IMPACT

By hosting the solution on a Django-based web framework, the system ensures high scalability and security. The integration of a SQLite database allows for the maintenance of longitudinal pet health records, which provides veterinarians with historical data during consultations—a feature currently missing in the fragmented veterinary landscape.

V. FLOWCHART



VI. CONCLUSION

The PetDoc project moves beyond the fragmented nature of current veterinary research by providing a cohesive, intelligent, and accessible platform. By solving the critical issues of delayed diagnosis and the logistical inconvenience of physical product procurement, this solution represents a significant advancement in smart veterinary care. Rather than functioning as a set of isolated tools, the system creates a seamless "discovery-to-recovery" pipeline. It empowers pet owners with immediate AI-driven insights, while simultaneously providing the e-commerce infrastructure necessary to act on those insights. This integration not only reduces the time-to-treatment for domestic animals but also establishes a new standard for user-centered digital health, proving that advanced machine learning can be successfully transitioned from theoretical research into a practical, life-saving application.